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May 19, 1947

FEDERAL SPECIFICATION

# MAGNIFIER AND REDUCING GLASS

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers the requirements for magnifiers, and reducing glasses of the types used by agencies of the Federal Government.

**1.2 Classification**

**1.2.1** *Types, classes, styles, and grades.* Magnifiers and reducing glasses shall be of the following types, classes, styles, and grades:

Type I—Reading Glasses.
- Class A—Round.
- Class B—Rectangular.
- Class C—Rectangular, illuminated.
- Class D—Rectangular, adjustable stand.
- Style 1—Illuminated.

Type II—Double Lens, Round.
- Class A—Hand held.
- Class B—Fixed focus, illuminated.
- Style 1—Battery.
- Style 2—115 volts.

Type III—Folding Pocket.
- Class A—Single lens.
- Class B—Double lens.
- Class C—Triple lens.
- Class D—Highly corrected lens (Hastings).

Type IV—Watchmakers Loupe.
- Class A—Single lens.
- Style 1—Without headband.
- Style 2—With headband.
- Grade A—Filtered.
- Class B—Double lens.
- Style 1—Without headband.
- Style 2—With headband.
- Grade A—Filtered.
- Class C—Hastings Triplet.

Type V—Eyeglass Loupes.
- Class A—Double lens.
- Class B—Filtered.

Type VI—Fingerprint Magnifier.

Type VII—Linen Tester.
- Class A—Small.
- Class B—Large.

Type VIII—General Purpose.
- Class A—Hastings, illuminated.
- Style 1—Battery.
- Style 2—115 volts.
- Class B—Flashlite type—illuminated.
- Style 1—Battery.
- Style 2—115 volt.
- Class C—Coddington, illuminated.

Type IX—Reducing Glass.

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

**2.1** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

*Federal Specifications:*

O-E-760—Ethyl Alcohol (Ethanol); Denatured Alcohol: and Proprietary Solvent.

KK-L-154—Leather, Cattlehide (Bag and Case).

QQ-S-763—Steel Bars, Shapes and Forgings—Corrosion Resistant.

QQ-S-766—Steel Plates, Sheets and Strip—Corrision Resistant.

NNN-P-40—Paper, Lens Tissue.

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

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PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-T-76—Tape, Pressure Sensitive, Adhesive, Paper (for Carton Sealing).

*Federal Standards:*

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

*Military Specifications:*

MIL-P-116—Preservation, Methods of.
MIL-G-174—Glass, Optical.
MIL-P-10408—Plastic Cellulose Acetate Molding Material and Molded Parts.

*Military Standards:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-171—Finishing of Metal and Wood Surfaces.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Design.** The figures in section 6 for each type of magnifier are for information purposes only and with the exception of lens requirements are not restrictive as to design characteristics.

**3.2 Material.**

3.2.1 *Glass.* The glass used in making the lens shall be of ophthalmic quality crown glass free from striae or green tint when viewed through the faces of the lens.

3.2.2. *Plastics.* Plastic material used in the magnifiers and reducing glasses shall be of the types best suited for optical instruments and shall not warp or change shape after prolonged use.

3.2.3 *Metals.* Metal parts shall be made of lightweight corrosion resistant steel.

**3.3 Lenses.** Lenses shall be accurately ground and polished, free from bubbles, seeds, digs and scratches. There shall be no evidence of blur, waves, grey or stain through any part of the lens more than 1/16 of an inch from the edge. The magnifying powers, focal length and dimensions of the lens shall be as specified in the applicable tables. A tolerance of plus or minus 10% is permitted. A tolerance of plus or minus 1/16 of an inch is permitted on all dimensions not otherwise limited.

**3.4 Identification.** The name or trademark of the supplier shall be permanently inscribed on each magnifier.

**3.5 Detail requirements.**

3.5.1 *Type I, class A, round reading glass (figure 1).*

3.5.1.1 *Design.* The round reading glass shall consist of a double convex lens mounted securely in a rim permanently attached to a handle.

3.5.1.2 *Rim.* The rim shall be of sufficient strength to support the lens and wide enough so that the lens shall not contact a flat surface.

3.5.1.3 *Handle.* The handle shall be permanently attached to the rim. The handle shall be of such shape as to be comfortably held by the user.

3.5.1.4 *Lens requirements.* The lens diameter, magnifying power and focal length shall be as specified in table I.

3.5.2 *Type I, class B, rectangular reading glass (figure 2).* The rectangular reading glass shall consist of a double convex lens rectangular in shape mounted in a plastic or metal frame of sufficient strength and rigidity to hold



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the lens firmly in place. The frame shall have a handle extending from a corner of the frame at an angle of 45 to 60 degrees to the larger dimension of the lens. The handle shall be of sufficient length to permit resting arm in a natural position when using the magnifier.

TABLE I.—*Round reading glasses*

| Magnifying power | Focal length | Lens diameter |
|---|---|---|
| | *Inches* | *Inches* |
| 8X | 13 | 5 |
| 1.0X | 10 | 4 |
| 1.3X | 8 | 3¼ |
| 1.7X | 6 | 2½ |
| 2.0X | 5 | 2 |

3.5.2.1 *Lens requirements.* The lens dimensions, magnifying power and focal length of the rectangular reading glass shall be as specified in table II.

TABLE II.—*Rectangular reading glasses*

| Magnifying power | Focal length | Lens dimensions |
|---|---|---|
| | *Inches* | *Inches* |
| 1.1X | 9 | 2 x 3⅜ |
| 1.3X | 8 | 1½ x 2⅜ |

3.5.3 *Type I, class C, rectangular, illuminated (figure 3).* The rectangular illuminated reader shall be optically the same as the class B reader and shall be provided with an integral illumination system consisting of a 110-volt line cord, an "OFF" and "ON" switch, and a 6-watt lamp mounted in the handle with a reflector that will direct the light on the area being examined.

3.5.4 *Type I, class D, rectangular, adjustable stand (figure 4).* The rectangular reader with adjustable stand shall be optically the same as the class B reader. The frame in which the lens is mounted shall be connected to an arm which is attached to a weighted base. The attachment between the frame and arm shall provide adjustment in both elevation and rotation. The attachment between the arm and base shall provide adjustment in elevation.

3.5.4.1 *Type I, class D, style 1, rectangular, adjustable stand, illuminated (figure 5).* This style of reader shall be optically the same as the class B reader and shall be provided with an illumination system consisting of a 110-volt line cord, an "ON" and "OFF" switch, a 15-watt lamp with a reflector that will direct the light on the area being examined.

3.5.5 *Type II, double lens magnifiers (figure 6).*

3.5.5.1 *Type II, class A.* The class A magnifier shall consist of two plano convex lenses permanently mounted in a plastic or hard rubber lens mount which shall be of sufficient strength and thickness to protect the lens from damage. The magnifier shall have a magnifying power of 3.5 and a minimum diameter of 1⅝ inches.

3.5.5.2 *Type II, class B, fixed focus, illuminated (figure 7).* The class B magnifier shall consist of two plano convex lenses permanently mounted in a metal housing. The metal housing shall position the lenses so that a flat surface upon which the housing rests shall be in focus. The magnifier shall have a magnifying power of 3.0 and a minimum diameter of 1⅝ inches. The magnifier housing shall be provided with a 6-watt lamp, and an "ON" and "OFF" switch and reflector that will direct the light on the area being examined.

3.5.5.2.1 *Type II, class B, style 1.* The style 1 magnifier shall be powered by two self-contained flashlight batteries.

3.5.5.2.2 *Type II, class B, style 2.* The style 2 magnifier shall be equipped with a line cord to permit 110-volt ac operation.

3.5.6 *Type III, classes A, B, and C, folding pocket magnifiers (figure 8).* The folding pocket magnifiers shall consist of a single, double, or triple lens mounted in a plastic frame attached to a plastic or leather case by a rivet on which the mounted lens shall pivot.

3.5.6.1 *Lens requirements.* The lens diameter, magnifying power and focal length of the type II magnifiers shall be as specified in table III.

3.5.6.2 *Type III, class D, folding pocket magnifiers, highly corrected, Hastings triplet (figure 9).* The Hastings Triplet magnifier shall consist of three round lenses cemented to-

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TABLE III.—*Folding pocket magnifiers*

| Class | Magnifying power | Focal length | Lens diameter |
|---|---|---|---|
| | | *Inches* | *Inches* |
| Class A* 1 lens | 2 × | 5 | 2. |
| | 3 × | 3 | 1. 5. |
| | 4 × | 2. 5 | 1. |
| Class B* 2 lenses | 3 ×–3.5=6 × | 3⅜–1¼ | 1¼–1½. |
| | 4 ×–5 ×=9 × | 2½–1⅝ | 1–1. |
| Class C 3 lenses | 5 ×–7 ×–10 ×= 20 ×. | 2–½ | ¾–⅝–½. |

NOTE: *Class A available in 3 choices of magnification; class B available in 2 choices of magnification.

gether, highly corrected for spherical and chromatic aberration. The lens shall be mounted in a plastic or chrome-plated metal frame attached to a chrome-plated metal case by a rivet on which the lens swings in or out.

3.5.6.3 *Lens requirements.* The lens diameter, magnifying power and focal length of the class D magnifiers shall be as specified in table IV.

TABLE IV.—*Class D magnifier—Hastings triplet*

| Magnifying power | Focal length | Lens diameter |
|---|---|---|
| | *Inches* | *Inch* |
| 7× | 1⅛ | ¾ |
| 10× | 1⅛ | ⅝ |
| 14× | ¹³⁄₁₆ | ½ |
| 20× | ½ | ⅜ |

3.5.7 *Type IV, watchmakers loupes (figure 10).*

3.5.7.1 *Type IV, class A, style 1.* The class A, style 1 loupe shall consist of a single 1-inch-diameter lens permanently mounted in one end of the black plastic loupe body. The eye end of the loupe body shall be so shaped that it can be held by the facial muscles of the eye.

3.5.7.2 *Type IV, class A, style 2.* The class A, style 2 loupe shall be the same as style 1, except that the style 2 loupe shall be provided with a spring steel headband that clips onto the loupe body and goes around the head.

3.5.7.3 *Type IV, class B, style 1.* The class B, style 1 loupe shall consist of two lenses, one permanent and one removable, which have two different magnifying powers. The 1-inch-diameter lower power lens shall be permanently mounted in one end of the plastic loupe body. The half-inch-diameter lens shall be permanently mounted in a threaded plastic collar that screws on the lens end of the loupe body for greater magnification.

3.5.7.4 *Type IV, class B, style 2.* The style 2 loupe shall be the same as the style 1, except that the style 2 loupe shall be provided with a spring steel headband that clips onto the loupe body and goes around the user's head.

3.5.7.5 *Type IV, classes A and B, styles 1 and 2, grade A.* The grade A loupes shall be the same as those specified in 3.5.7.1 and 3.5.7.3 except that the 1-inch lens shall be tinted to provide absorption when used under adverse lighting conditions.

3.5.7.6 *Type IV, class C, Hastings triplet.* The class C Hastings triplet loupe shall consist of a ⅝-inch-diameter, 10-power, Hastings triplet lens mounted in one end of the plastic loupe body.

3.5.7.7 *Lens requirements.* The lens diameter, magnifying power and focal length of the type IV magnifiers shall be as specified in table V.

3.5.8 *Type V, eyeglass loupes (figure 11).*

3.5.8.1 *Type V, class A, double lens.* The class A eyeglass loupe shall consist of two 1-inch lens mounted in metal rims of the split-ring type which are attached to a metal holder that clamps to the frame of standard eyeglasses. The eyeglass loupes shall be able to rotate independently about the holder.



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3.5.8.2 *Type V, class B, double, filtered lens.* The class B double lens eyeglass loupe shall be the same as specified in 3.5.8.1 except that the lens closest to the user's eye shall be tinted to provide absorption when used under adverse lighting conditions.

3.5.8.3 *Lens requirements.* The lens diameter, magnifying power and focal length of the type V magnifiers shall be as specified in table VI.

3.5.9 *Type VI, fingerprint magnifier (figure 12).* The fingerprint magnifier shall consist of two optically ground and polished lenses 1 9/16 inches in diameter with a total magnification of 4.5× power. The lens shall be mounted in a plastic frame which is mounted in a metal horseshoe type base. The upper part of the base shall have a split arm that is joined by a knurled screw that allows for focusing of the lens. The lower part of the base shall allow for insertion of a disc for fingerprint classification.

3.5.10 *Type VII, classes A and B, linen tester (figure 13).* The linen tester shall consist of three metal segments hinged together in such a way that when in the open position the middle segment shall hold the base segment and lens segment in relative parallel positions. The base segment shall have a square opening which shall have graduated divisions on the surface. The graduations shall have an accuracy of 0.01 of an inch. The middle segment shall have a large diameter hole opening to permit maximum light on the material being examined. The double lens shall be permanently mounted in the lens segment. The axis of the lens and the opening in the base segment shall be alined. The lens shall be in focus with the material held in a plane against the bottom of the base, when the linen tester is in the open position.

3.5.10.1 *Lens requirements.* The lens diameter, magnifying power, base openings and scale divisions shall be as specified in table VII.

TABLE V.—*Watchmakers loupes*

| Class | Magnifying power | Focal length | Lens diameter |
|---|---|---|---|
| | | *Inches* | *Inch* |
| A—Single lens | 2.5× | 4 | 1 |
| | 5× | 2 | 1 |
| B—Two lenses | 4× and 7× | 2½ and 1½ | ⅞ and ½ |
| | 8× and 18× | 1¼ and ⅝ | ⅞ and ½ |
| C—Hastings triplet | 10× | 1 | ⅝ |

TABLE VI.—*Eyeglass loupes*

| Class | Magnifying power | Focal length | Lens diameter |
|---|---|---|---|
| | | *Inches* | *Inch* |
| A—Two lenses | 4× and 7× | 2½–1½ | 1 and 1. |

TABLE VII.—*Linen tester*

| Class | Magnifying power | Diameter of lens | Base opening | Division of scale |
|---|---|---|---|---|
| | | *Inches* | *Inch* | *Inch* |
| A—Small | 7. 5× | ⅝ | ½x½ | 1/16 |
| B—Large | 5. 0× | 1⅛ | 1x1 | ¼ |

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3.5.10.2 *Case.* The linen tester shall be supplied with a leather case that will accommodate the magnifier when it is in the folded position.

3.5.11 *Type VIII, general purpose magnifiers (figures 14, 15, and 16).*

3.5.11.1 *Class A, Hastings, illuminated (figure 14).* The class A magnifier shall consist of a 5-power Hastings triple aplantic lens mounted in a metal frame which is inserted into a metal housing, which has built in illumination.

3.5.11.1.1 *Class A, style 1.* The class A, style 1 magnifier shall be illuminated by battery.

3.5.11.1.2 *Class A, style 2.* The class A, style 2 magnifier shall be equipped with a line cord to permit 110-volt ac operation.

3.5.11.2 *Class B, flashlight type, illuminated (figure 15).* The class B magnifier shall consist of a 5- or 7-power double lens mounted in a metal frame of a flashlight design, equipped with an ON and OFF switch.

3.5.11.2.1 *Class B, style 1.* The class B, style 1 magnifier shall be illuminated by a battery.

3.5.11.2.2 *Class B, style 2.* The class B, style 2 magnifier shall be equipped with a line cord to permit 110-volt ac operation.

3.5.11.3 *Class C, Coddington, illuminated (figure 16).* The class C magnifier shall consist of a Coddington 10-power lens mounted in a plastic frame, which is threaded into a metal fountain pen type holder, with built-in illumination powered by batteries. The holder shall have an ON and OFF switch and clip for attaching to the user's pocket.

3.5.11.4 *Lens requirements.* The lens diameter, magnifying power and focal length shall be as specified in table VIII.

TABLE VIII.—*General purpose magnifiers*

| Class | Magnifying power | Focal length | Lens diameter |
|---|---|---|---|
| | | *Inches* | *Inches* |
| A—Hastings | 5× | 2 3/16 | 1 1/16 |
| B—Double lens | 5× | 2 3/16 | 2 |
| C—Coddington | 10× | 1 1/8 | 3/4 |

3.5.12 *Type IX reducing glass (figure 1).* The type IX reducing glass shall comply with the requirements of 3.5.1 except that the lens shall be concave instead of convex. The lens shall have a reducing power of minus 1.4×, a diameter of 3 inches with a 7-inch focal length.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

**4.2 Sampling.** Sampling shall be in accordance with MIL-STD-105.

**4.3 Material.** Materials shall be in accordance with pertinent specifications forming part of this specification.

**4.4 Glass.** Glass material shall be visually inspected for compliance with 3.2.1 and MIL-G-174.

**4.5 Plastic.** Plastic, molded parts, shall be inspected for conformance of the requirements of 3.2.2 and MIL-P-10408.

**4.6 Metals.** Metal components manufactured from corrosion resisting steel shall be inspected to the requirements of 3.2.3 and QQ-S-763 or QQ-S-766. Components manufactured from corrosive metals must have a corrosive resisting protective finish applied and shall be inspected to the requirements of MIL-STD-171.

**4.7 Lenses.** The optical lens components shall be inspected for conformance to the requirements of 3.3.

**4.8 Detail requirements.**

4.8.1 *Type 1, class A.*

4.8.1.1 *Design, rim, handle.* This item shall be visually inspected to be certain that it conforms to the requirements specified in 3.5.1.1, 3.5.1.2 and 3.5.1.3.

4.8.1.2 *Lens requirements.* Optical components shall be inspected for conformance to table I and 4.7, using standard measuring equip-

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ment to determine values as required by 3.5.1.4.

**4.9 Type I, class B.** This item shall be visually inspected to be certain that it conforms to the requirements specified in 3.5.2.

4.9.1 *Lens requirements.* Optical components shall be inspected for conformance to table II and 4.7 using standard measuring equipment to determine values as required by 3.5.2.1.

**4.10 Type I, class C.** This item shall be visually inspected to be certain that it conforms to the requirements specified in 3.5.3.

**4.11 Type I, class D, styles 1 and 2.** This item shall be visually inspected to be certain that it conforms to the requirements specified in 3.5.4 and 3.5.4.1.

**4.12 Type II, double lens magnifier.**

4.12.1 *Type II, class A.* Optical components of this item shall be inspected for conformance to 4.7 using standard measuring equipment to determine values specified as required by 3.5.5.1.

4.12.2 *Type II, class B.* This item shall be visually inspected to be certain that it conforms to the requirements specified in 3.5.5.2. The optical components shall be inspected for conformance to 4.7 using standard measuring equipment to determine values specified.

4.12.3 *Type II, class B, styles 1 and 2.* Items in this category shall be visually inspected to be certain that they conform to the requirements specified by 3.5.5.2.1 and 3.5.5.2.2.

**4.13 Type III, folding pocket magnifier.** This item shall be visually inspected to be certain that it conforms to the requirements specified in 3.5.6.

4.13.1 *Lens requirements.* Optical components shall be inspected for conformance to table III and 4.7 using standard measuring equipment to determine values specified by 3.5.6.1.

**4.14 Type III, class D.** Items in this category shall be visually inspected to be certain that they conform to requirements specified in 3.5.6.2.

4.14.1 *Lens requirements.* Optical components shall be inspected for conformance to table IV and 4.7 using standard measuring equipment to determine values specified in 3.5.6.3.

**4.15 Type IV, watchmakers loupes, all classes and styles.** Items in this category shall be visually inspected to be certain that they conform to the requirements specified by 3.5.7.1, 3.5.7.2, 3.5.7.3, 3.5.7.4, 3.5.7.5 and 3.5.7.6.

4.15.1 *Lens requirements.* Optical components shall be inspected for conformance to table V and 4.7 using standard measuring equipment to determine values specified in 3.5.7.7.

**4.16 Type V, eyeglass loupes, all classes and styles.** Items in this category shall be visually inspected to be certain that they conform to the requirements specified by 3.5.8.1 and 3.5.8.2.

4.16.1 *Lens requirements.* Optical components shall be inspected for conformance to table VI and 4.7 using standard measuring equipment to determine values specified in 3.5.8.3.

**4.17 Type VI.** This item shall be visually inspected to be certain that it conforms to the requirements specified. Optical components shall be inspected for conformance to 4.7 using standard measuring equipment to determine values specified by 3.5.9.

**4.18 Type VII, classes A and B.** Items in this category shall be visually inspected to be certain that they conform to the requirements specified in 3.5.10.

4.18.1 *Lens requirements.* Optical components shall be inspected for conformance to table VII and 4.7 using standard measuring equipment to determine values specified by 3.5.10.1.

4.18.2 *Case.* This item shall be visually inspected to be certain that it conforms to the requirements specified by 3.5.10.2.

**4.19 Type VIII, all classes and types.** Items in this category shall be visually inspected to be certain that they conform to the requirements specified by 3.5.11.1, 3.5.11.1.1, 3.5.11.1.2, 3.5.11.2 and 3.5.11.3.

4.19.1 *Lens requirements.* Optical components shall be inspected for conformance to table VIII and 4.7 using standard measuring equipment to determine values specified in 3.5.11.4.

**4.20 Type IX.** This item shall be visually inspected to be certain that it conforms to the

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requirements specified in 3.5.1.1, 3.5.1.2 and 3.5.1.3.

4.20.1 *Lens.* The lens shall be inspected for conformance to 3.3 using standard measuring equipment to determine values required by 3.5.12.

4.21 **Preservation, packaging, packing, and marking.** Magnifiers shall be inspected to sampling plan of MIL–STD–105 for conformance to the requirements specified in section 5 as applicable to level of packaging and packing ordered.

4.22 **Workmanship.** Magnifiers shall be visually inspected to ascertain that protective finishes are free of chipping, flaking, peeling and that no heavy flashing or burrs are present on molded and cast parts.

## 5. PREPARATION FOR DELIVERY

(The levels of preservation, packaging and packing protection used are defined in Fed. Std., No. 102 for civil agencies only.)

5.1 **Preservation and packaging.** Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The lens of the magnifiers shall be cleaned with alcohol conforming to O–E–760.

5.1.1.2 *Drying.* The magnifier shall be dried by any applicable method of MIL–P–116 and wrapped in lens tissue paper conforming to NNN–P–40.

5.1.1.3 *Unit packaging.* Each magnifier shall be individually packaged method IA–8 of MIL–P–116.

5.1.1.4 *Intermediate packaging.* Ten unit packages, of the same type of magnifier, shall be further packaged in a close fitting fiberboard box conforming to class 2, W5c of PPP–B–636. The box shall be closed in accordance with the appendix to the box specification, using tape conforming to PPP–T–76.

5.1.2 *Level C.* The magnifiers shall be packaged in accordance with the manufacturer's commercial practice which shall afford adequate protection against corrosion, deterioration, and physical damage from the supply source to the first receiving activity for immediate use.

5.2 **Packing.**

5.2.1 *Level A.* Magnifiers of the same type packaged as specified in 5.1.1 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box or nailed wood box conforming to PPP–B–585, class 3; PPP–B–591 (overseas type); PPP–B–601, style A or B; PPP–B–621, class 2, respectively. Closure and strapping shall be as specified in the appendix of the applicable box specification. The gross weight shall not exceed 200 pounds.

5.2.2 *Level B.* Magnifiers of the same type packaged as specified in 5.1.1 shall be packed in a wirebound wood box, wood-cleated fiberboard box, wood-cleated plywood box or a nailed wood box conforming to PPP–B–585, class 1; PPP–B–591 (domestic type); PPP–B–601 (domestic type); PPP–B–621, class 1, respectively. The gross weight of the boxes shall not exceed the weight limitation of the applicable box specification.

5.2.3 *Level C.* The magnifiers shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Military.* Interior packages and exterior shipping containers shall be marked in accordance with MIL–STD–129.

5.3.2 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

## 6. NOTES

6.1 **Intended use.** Magnifiers covered by this specification are used to make an object appear larger and to increase its apparent dimensions so that its physical nature may be studied in greater detail. Reducing glasses are used to make an object appear smaller.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.



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(*b*) Quantity, type, class, style of magnifiers, and reducing glasses desired (1.2.1).
(*c*) Whether cases for type I and II magnifiers are required (1.2.1).
(*d*) Responsibility for inspection (4.1).
(*e*) Selection of applicable levels of packaging and packing required (5.1).

6.3 This specification does not include every type, class, grade, style, etc., of magnifiers which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If for any particular purpose samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

*Rail:*

Glasses, magnifying or reading, other than eyeglasses or spectacles.
Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

*Motor:*

Optical goods, not otherwise indexed.
Truckload minimum weight 10,000 pounds—subject to Rule 115, National Motor Freight Classification.

**Military Interests:**
**Army—MU.**
**Navy—BuShips.**
**Air Force—MAA.**

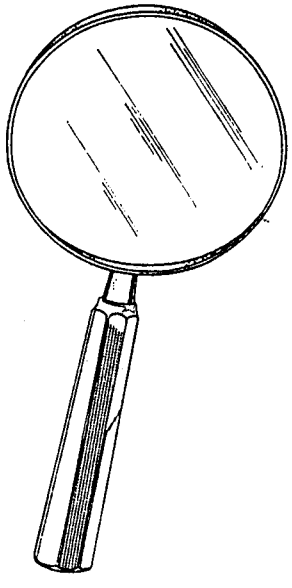


FIGURE 1.—*Type I, class A, round reading glass.*

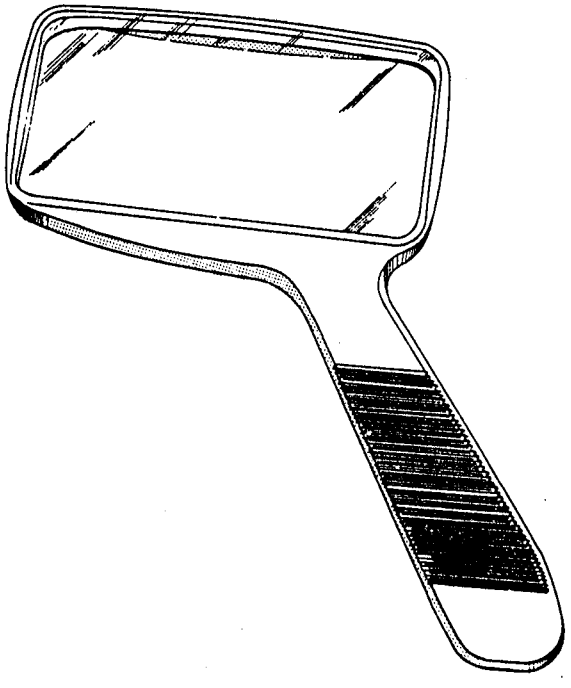


FIGURE 2.—*Type I, class B, rectangular reading glass.*

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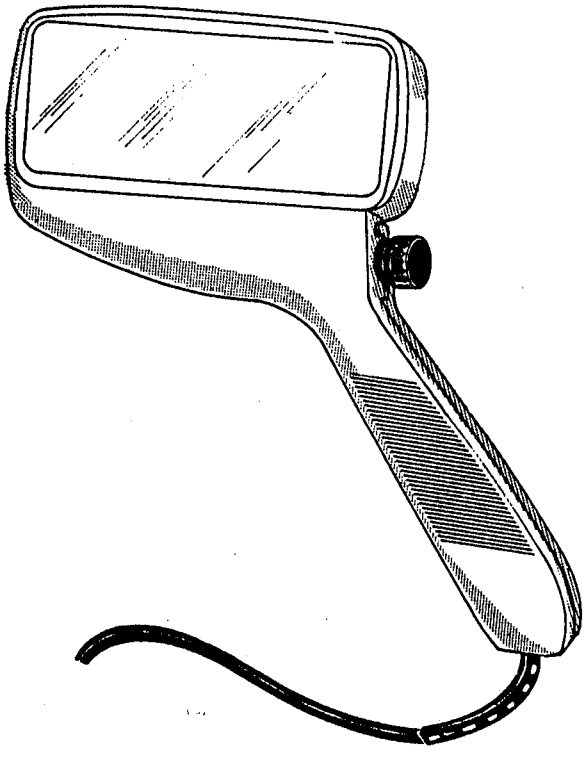


FIGURE 3.—*Type I, class C, rectangular, illuminated reading glass.*

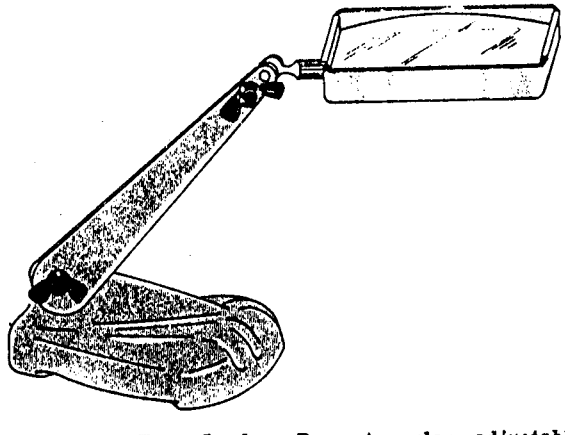


FIGURE 4.—*Type I, class D, rectangular, adjustable stand.*

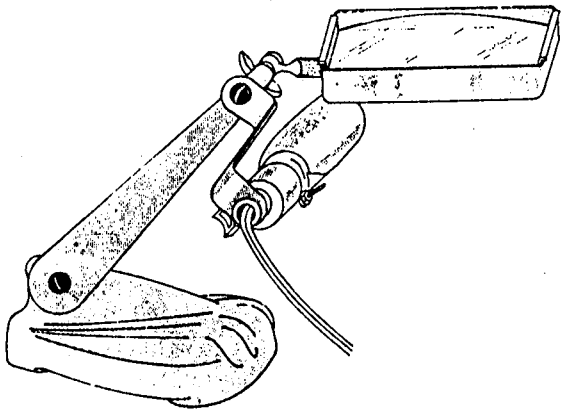


FIGURE 5.—*Type I, class D, style 1, rectangular, adjustable stand, illuminated.*

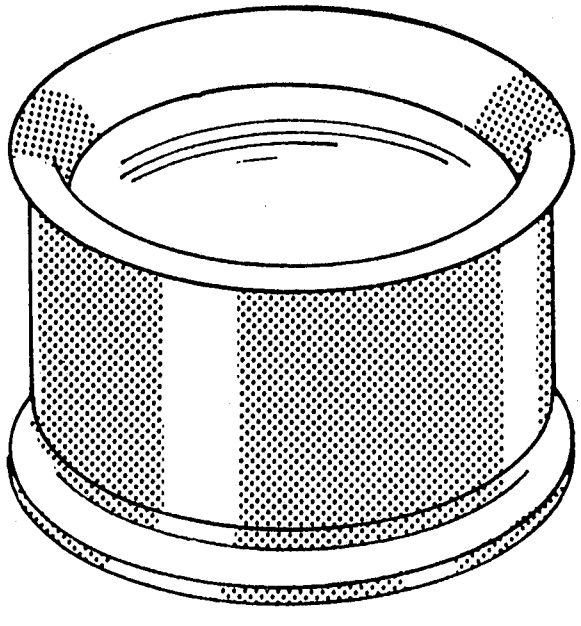


FIGURE 6.—*Type II, class A, double lens, round magnifier, hand held.*



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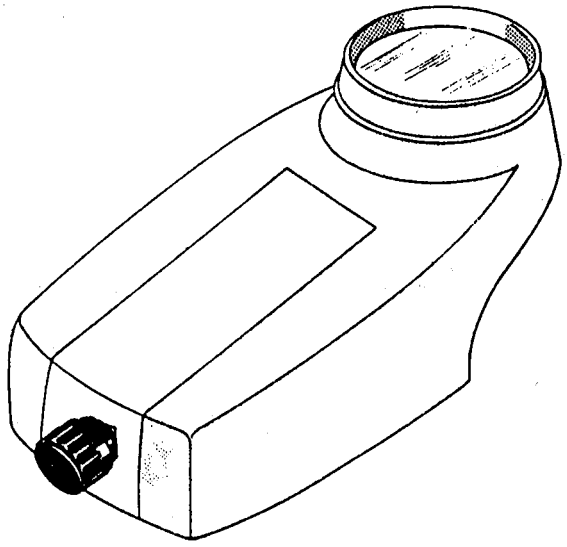


FIGURE 7.—*Type II, class B, double lens, round magnifier, fixed focus, illuminated.*

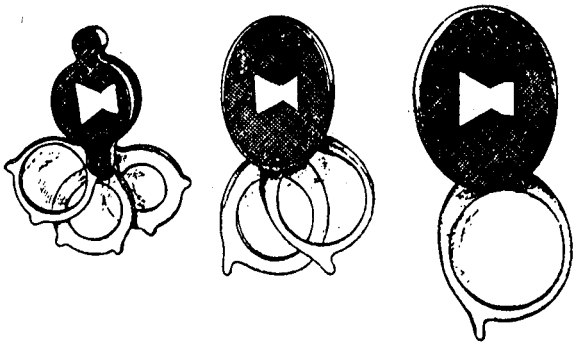


FIGURE 8.—*Type III, folding pocket magnifier:*
*Class A, single lens.*
*Class B, double lens.*
*Class C, triple lens.*

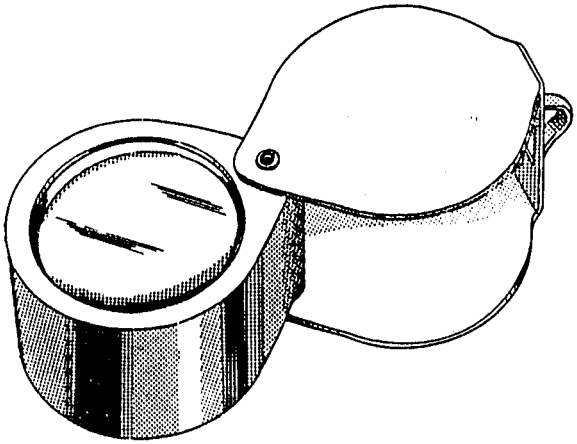


FIGURE 9.—*Type III, class D, folding pocket magnifier, highly corrected lens.*

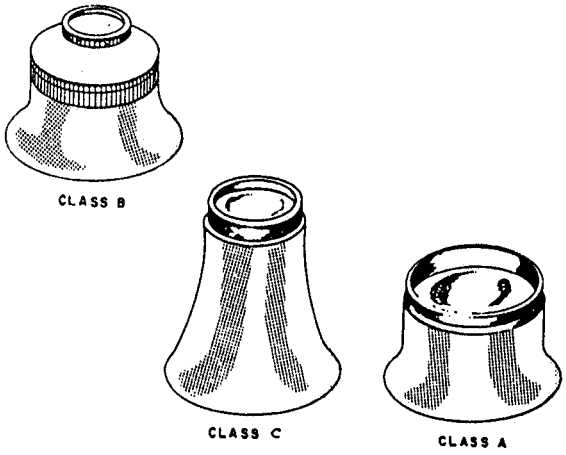



FIGURE 10.—*Type IV.—Watchmakers loupes:*
*Class A, single lens.*
*Class B, double lens.*
*Class C, Hastings triplet.*

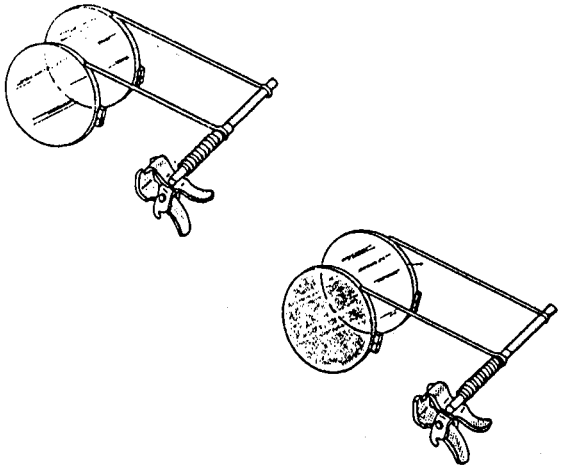


FIGURE 11.—*Type V, eyeglass loupes:*
*Class A, double lens.*
*Class B, double, filtered lens.*

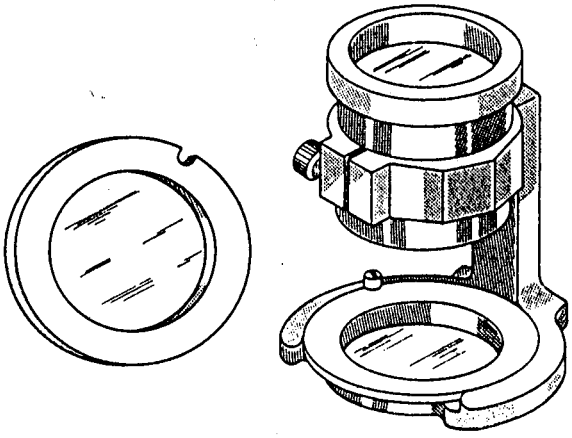


FIGURE 12.—*Type VI, fingerprint magnifier.*

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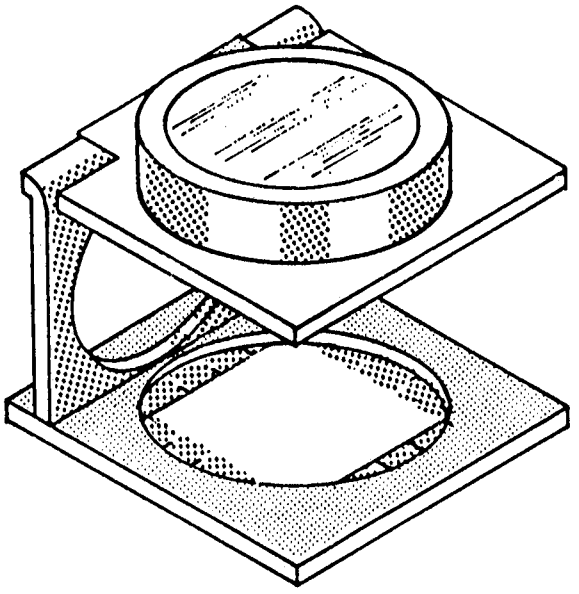


FIGURE 13.—*Type VII, linen tester.*

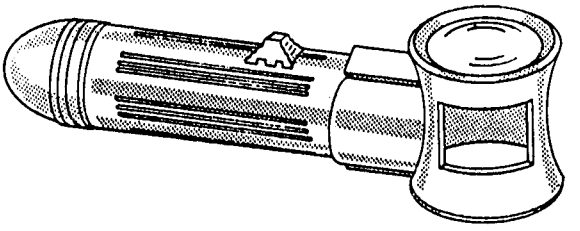


FIGURE 15.—*Type VIII, general purpose magnifier, class B, 5-power flashlite type, illuminated.*

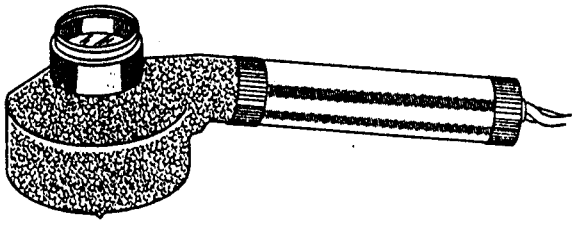


FIGURE 14.—*Type VIII, general purpose magnifier, class A, Hastings, illuminated.*

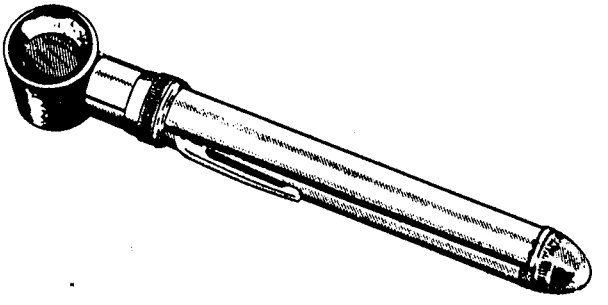


FIGURE 16.—*Type VIII, general purpose magnifier, class C, Coddington, illuminated.*

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